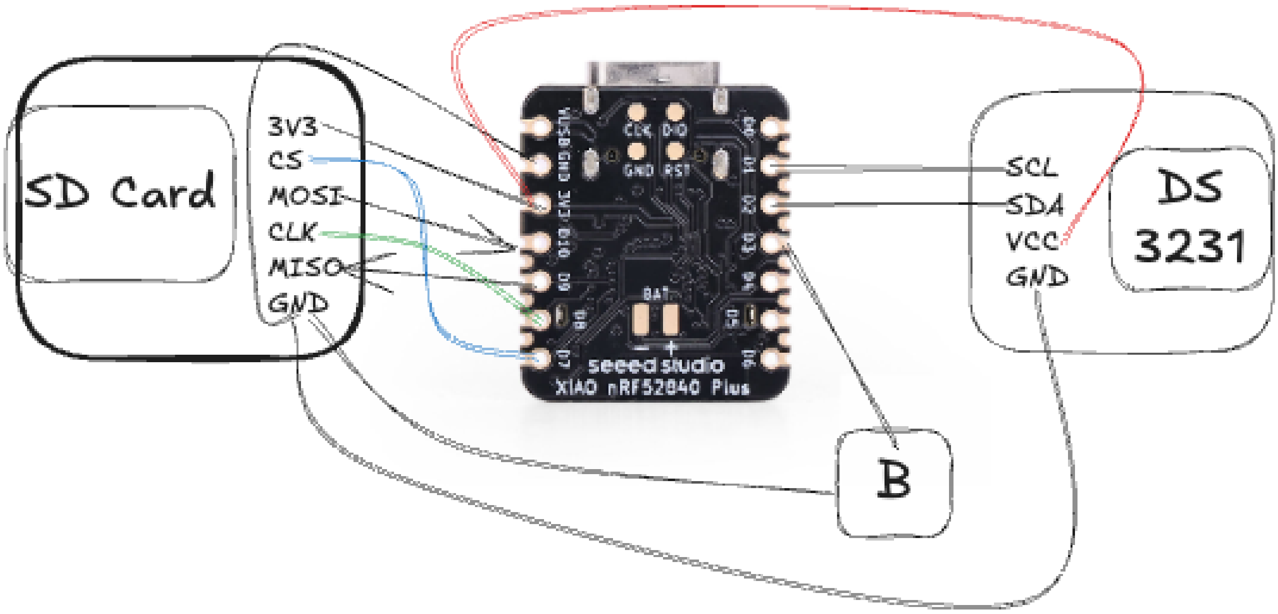
README

📟 NAPPNU Motion Data Recorder

Bluetooth-enabled Motion Data Logger built using the **Seeed XIAO nRF52840 Sense**.

🛠️ Hardware Components



Component	Quantity
Seeed XIAO nRF52840 Sense	1
DS3231 RTC Module	1
microSD Card Module	1
microSD Card (FAT32)	1
Push Button	1
Li-Po Battery (3.7 V)	1
USB Type-C Cable	1

📁 Project Structure

```
main-folder/
│
├──Mobile Application/
│   └──motion-logger
│           └──files.....
│
├──Code-Hardware/
│  └──Data_Logger.ino
│
├──PCB Design/
│  └──KiCAD.zip
│
├──Python-Script/
│  └──connect.py
│
└──README files
```

💻 Software Requirements

Firmware Development

- Arduino IDE 2.x
- Seeed nRF52 Board Package

Required Arduino Libraries

- Adafruit Bluefruit nRF52 Libraries
- LSM6DS3
- SdFat
- RTClib
- Wire
- SPI
- EEPROM

📦 Project Files

All required project files are available in the shared Google Drive folder below.

> 📁 **Download Project Files**
> https://drive.google.com/drive/folders/1yOiiscPsLclZifj_FWon-T-I5wJ6or7X?usp=drive_link

The folder contains:

- 📱 Pre-built Android APK
- 📱 Flutter source code

📱 Option 1 — Install the Pre-built APK (Recommended)

1. Open the **Google Drive** folder above.
2. Navigate to:

```
Android-Application/
└──motion_logger/
  └──build/
    └──app/
      └──outputs/
        └──flutter-apk/
          └──app-release.apk
```

3. Download **app-release.apk** to your Android device.(click here)
4. Install the APK.

> **Note:** Enable **Install from Unknown Sources** if prompted.

🧑‍💻 Option 2 — Development Setup

Download the complete project from the Google Drive folder.

Flutter

```
cd motion_logger

flutter pub get

flutter run
```

Generate a release APK:

```
flutter build apk --release
```

The generated APK will be available at:

```
build/app/outputs/flutter-apk/app-release.apk
```

Flutter Packages Used

- flutter_blue_plus
- permission_handler
- file_picker
- shared_preferences
- url_launcher
- path_provider

Desktop Client

- Python 3.11+
- Bleak

```
pip install bleak
```

🚀 Setup Procedure

📱 Device Controls

Button Action	Function
🟢 Single Press	Start / Stop Logging
🔵 Double Press	Enable / Disable BLE
🔴 Long Press (1.5 s)	Enter Deep Sleep
🟡 Single Press (Wake)	Wake Device

Step 1 — Upload Firmware

Open

```
1. Install and run Arduino ide with the Required arduino libraries(mentioned above)
2. Download Code-Hardware/Data_Logger.ino from drive
```

Select

```
Board
  ↓
Seeed XIAO nRF52840 Sense
```

Then click **Upload.**

Step 2 — Prepare the SD Card

- Format as **FAT32**
- Insert into the SD card module

📱 Step 3 — Client Applications

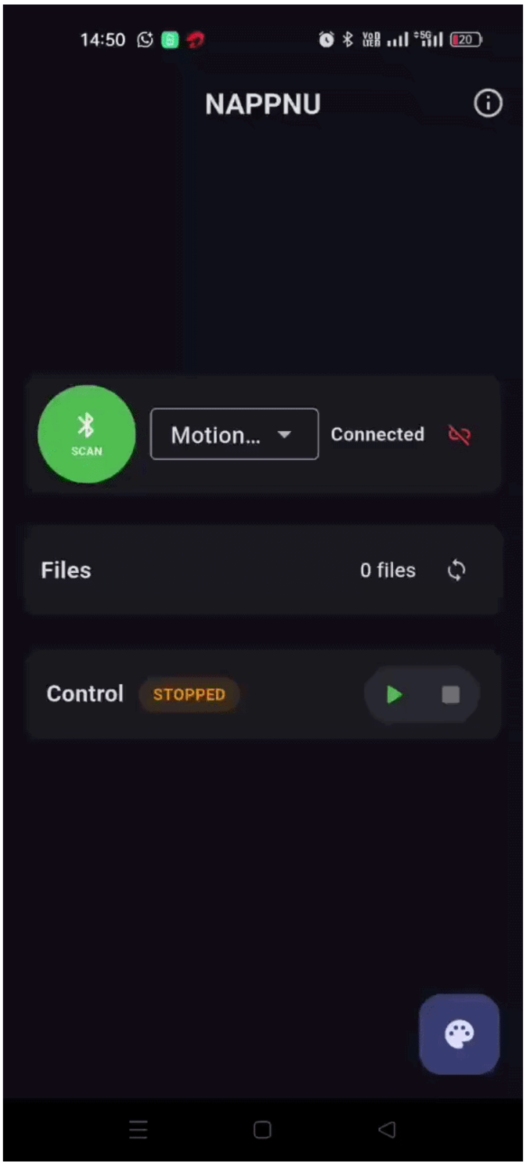
After building the firmware and mobile application, the Motion Data Logger can be controlled using **either** the Android application **or** the Desktop Python client.

> **Important**
>
> Only **one client can be connected to the logger at a time.**
>
> - ✅ Android Application **or**
> - ✅ Desktop Python Client
> - ❌ Both simultaneously

> **Note**
>
> Press the device button **twice** to enable **BLE advertising mode**. The logger advertises for **30 seconds**, allowing the Android application to discover and connect to it. If a connection is established within this interval, the device remains connected and is ready for normal operation until disconnected.

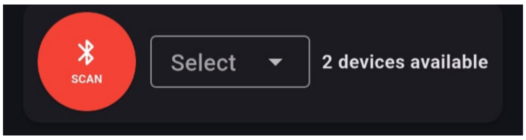
3.1.Android Application

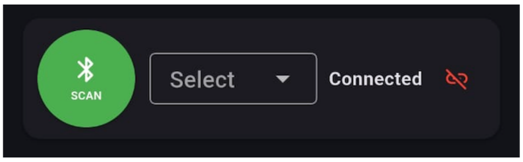
The Android application provides a graphical interface for controlling the logger and managing stored CSV files.



1. Device Discovery

- Scan for nearby MotionLogger devices.
- Select a device from the list.
- Connect using Bluetooth Low Energy (BLE).

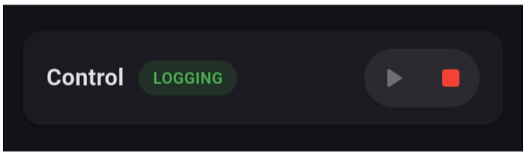




2. Logger Control

After connecting, the application allows you to:

- Start logging
- Stop logging
- View logger status



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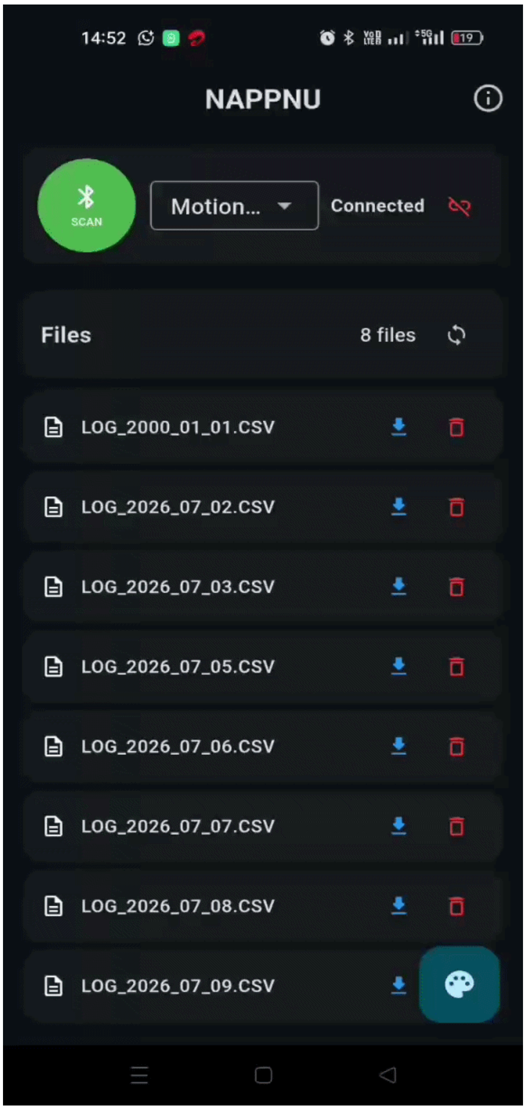
3. File Management

The application can communicate directly with the logger to:

- View all CSV files
- Download selected files
- Delete files from the SD card
- Refresh the file list



3.2.Desktop Client

A lightweight Python client is also provided for communicating with the logger from a computer.

The desktop client supports:

- Device discovery
- BLE connection
- Start / Stop logging
- View logger status
- List stored files
- Download files
- Delete files

Run the client->

```
cd path
python connect. py
```